Knowledge Representation

Lecture 13:

Default Logics II

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Representational Issues in DL

Example 13.1: Clever Students

Consider the following sentences:

- *Students are usually clever.*
- *Clever people usually make good career.*
- *Peter is a student.*

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Representation in DL:

$$A = \{Student(peter)\}$$

$$\Delta = \left\{ \frac{Student(x) : Clever(x)}{Clever(x)}, \frac{Clever(x) : MakesCareer(x)}{MakesCareer(x)} \right\}.$$

Example 13.1 (cont.)

The closure of $T = (A, \Delta)$:

$$A = \{Student(peter)\}$$

$$\Delta = \left\{ \begin{array}{l} \delta_1 = \dfrac{Student(peter) : Clever(peter)}{Clever(peter)}, \\[2ex] \delta_2 = \dfrac{Clever(peter) : MakesCareer(peter)}{MakesCareer(peter)} \end{array} \right\}.$$

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Application of δ_1 gives $Clever(peter)$. Then δ_2 is applicable, after its application we get $MakesCareer(peter)$.

Example 13.1 (cont.)

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Application of δ_1 gives $Clever(peter)$. Then δ_2 is applicable, after its application we get $MakesCareer(peter)$.

Here ***transitivity of defaults*** is desirable.

Example 13.2: Married Students

Consider the following sentences:

- *Mike is a student.*
- *Students are usually adults.*
- *Adults are usually married.*

Representation in DL:

$$A = \{Student(mike)\}$$

$$\Delta = \left\{ \frac{Student(x) : Adult(x)}{Adult(x)}, \frac{Adult(x) : IsMarried(x)}{IsMarried(x)} \right\}$$

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One can easily note that we get $IsMarried(mike)$. Moreover, for an arbitrary student we will obtain a similar conclusion!

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One can easily note that we get $IsMarried(mike)$. Moreover, for an arbitrary student we will obtain a similar conclusion!

Here ***transitivity of defaults*** is not desirable!

Example 13.2 (cont.)

In other to block undesirable conclusion, we can use a representation:

$$\Delta = \left\{ \begin{array}{c} \dfrac{Student(x) : Adult(x)}{Adult(x)} \,, \\ \dfrac{Adult(x) : IsMarried(x) \wedge \neg Student(x)}{IsMarried(x)} \end{array} \right\}.$$

In other words, we substitute the second rule by:

Typical adults are married unless they are students.

Only the first default is applicable and we get $Adult(mike)$. The previous conclusion $IsMarried(mike)$ cannot be derived anymore.

Example 13.3: Reasoning by Case

Consider the following sentences:

- *Tweety is a bird or a bee.*
- *Typical birds can fly.*
- *Typical bees can fly.*

Intuitively we feel that *Tweety can fly*.

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- *Tweety is a bird or a bee.*
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Intuitively we feel that *Tweety can fly*.

Representation in DL:

$$A = \{Bird(tweety) \vee Bee(tweety)\}$$

$$\Delta = \left\{ \frac{Bird(x) : CanFly(x)}{CanFly(x)}, \frac{Bee(x) : CanFly(x)}{CanFly(x)} \right\}$$

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$$\Delta = \left\{ \frac{Bird(x) : CanFly(x)}{CanFly(x)}, \frac{Bee(x) : CanFly(x)}{CanFly(x)} \right\}$$

None of two defaults is applicable wrt A, so $E = Th(A)$. Consequently, we cannot derive a natural conclusion "*Tweety can fly*".

Example 13.3 (cont.)

We can use the following representation:

$$\Delta = \left\{ \frac{: Bird(x) \to CanFly(x)}{Bird(x) \to CanFly(x)}, \; \frac{: Bee(x) \to CanFly(x)}{Bee(x) \to CanFly(x)} \right\}$$

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Now we get

$$Bird(tweety) \to CanFly(tweety)$$
$$Bee(tweety) \to CanFly(tweety),$$

or equivalently $Bird(tweety) \vee Bee(tweety) \to CanFly(tweety)$. From A we immediately get: $CanFly(tweety)$.

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- *Students are usually adults*.
- *Peter is a teenager*.

Intuitively, we want to conclude: *Peter is not a student*.

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$$A = \left\{ \begin{array}{l} Teenager(peter) \\ \forall x.\, Teenager(x) \rightarrow \neg Adult(x) \end{array} \right\}$$

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From A we conclude $\neg Adult(peter)$. Since $\delta(peter)$ is not applicable, we cannot derive the expected conclusion!

Example 13.4 (cont.)

We can use another representation:

$$A = \left\{ \begin{array}{l} Teenager(peter) \\ \forall x.\ Teenager(x) \rightarrow \neg Adult(x) \end{array} \right\}$$

$$\Delta = \left\{ \ \delta(x) = \frac{: Student(x) \rightarrow Adult(x)}{Student(x) \rightarrow Adult(x)} \ \right\}$$

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$$\Delta = \left\{ \delta(x) = \frac{: Student(x) \rightarrow Adult(x)}{Student(x) \rightarrow Adult(x)} \right\}$$

From A we get $\neg Adult(peter)$.
$\delta(peter)$ is applicable and after its application we obtain
$Student(peter) \rightarrow Adult(peter)$.
Hence we get $\neg Student(peter)$.

Algorithm for computing extensions

Auxiliary notations and notions

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- Let M be a set of models (of some set B of formulas – beliefs) and let J be the set of formulas (justifcations of these beliefs).

  A pair (M, J) is a ***(M, J)–pair*** iff for every $\varphi \in J$, $M \cap Mod(\varphi) \neq \emptyset$.

  E.g.,
  - $(\{m : m \models p \wedge q\}, \{r, \neg r\})$ is an MJ–pair
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- A default $\delta = \dfrac{\alpha : \beta_1, \ldots, \beta_n}{\gamma}$ is applicable wrt M iff
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Stability

Let (M, J) be an MJ–pair and let Δ be a set of closed defaults. (M, J) is called ***Δ–stable*** iff for every default $\delta = (\alpha : \beta_1, \ldots, \beta_n)/\gamma \in \Delta$ it holds:

if δ is applicable wrt M, then

- $M \cap Mod(\gamma) = M$
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In other words,

- either δ is applicable, but its application does not introduce new information,
- or δ it is not applicable (and is just skipped).

Mapping d_δ

Let M be a set of models and let J be a set of formulas. For simplicity we will consider defaults of the form $\delta = (\alpha : \beta)/\gamma$. For every δ define a mapping d_δ, which transforms a pair (M, J) into another pair (M', J') as follows:

$$d_\delta(M,J) = \begin{cases} (M \cap Mod(\alpha), J \cup \{\beta\}) & \text{iff } \delta \text{ is applicable wrt } M \\ & \text{and } (M,J) \text{ is an (M,J)–pair} \\ (M,J) & \text{iff } \delta \text{ is not applicable wrt } M \\ & \text{and } (M,J) \text{ is an (M,J)–pair} \\ (\emptyset, \mathcal{L}) & \text{otherwise.} \end{cases}$$

Example 13.5

Consider the following sets:

$$
\begin{aligned}
M &= \{m : m \models p \wedge q\} = \left\{ \begin{array}{l} \{p, q, r, s\} \\ \{p, q, r, \neg s\} \\ \{p, q, \neg r, s\} \\ \{p, q, \neg r, \neg s\} \end{array} \right\}, \\
J &= \{\neg r \wedge q, s\}, \\
\Delta &= \left\{ \delta_1 = \frac{p : q \wedge \neg r}{q}, \ \delta_2 = \frac{r : s}{q} \right\}.
\end{aligned}
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- $d_{\delta_1}(M, J) = (M, J)$ — δ_1 is applicable
- $d_{\delta_2}(M, J) = (M, J)$ — δ_2 is inapplicable.

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(M, J) is Δ–stable.

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$$M = \{m : m \models p \wedge q\} = \left\{ \begin{array}{l} \{p, q, r, s\} \\ \{p, q, r, \neg s\} \\ \{p, q, \neg r, s\} \\ \{p, q, \neg r, \neg s\} \end{array} \right\}, \quad J = \{q \wedge \neg r, r\}$$

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- $d_{\delta_2}(M, J) = \{m : m \models p \wedge q \wedge r\}, J) \neq (M, J)$.

Notice that $\varphi = q \wedge \neg r$ does not hold in $\{m : m \models p \wedge q \wedge r\}$, so $d_{\delta_2}(M, J)$ is not an MJ–pair.

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Notice that $\varphi = q \wedge \neg r$ does not hold in $\{m : m \models p \wedge q \wedge r\}$, so $d_{\delta_2}(M, J)$ is not an MJ–pair.

(M, J) is not Δ–stable.

Algorithm

- **Input:** A closed default theory $T = (A, \Delta)$
- **Output:** All extensions $E_1, \ldots, E_k$ of T.
  In fact, we will determine the set $M_1, \ldots, M_k$ of all models of extesions of T.

Algorithm (cont.)

Let $T(A, \Delta)$ and assume that $\Delta \neq \emptyset$.

(S1) Put $P := PERM(\Delta)$; $\mathfrak{M} := \emptyset$

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Let $T(A, \Delta)$ and assume that $\Delta \neq \emptyset$.

(S1) Put $P := PERM(\Delta)$; $\mathfrak{M} := \emptyset$

(S2) If $P = \emptyset$ then STOP $\Longrightarrow \mathfrak{M}$ semantically corresponds to the set of all extensions of T; otherwise $M := \text{Mod}(A)$; $J := \emptyset$;

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(S4) If $i > k$ then
if (M, J) is Δ–stable then $\mathfrak{M} := \mathfrak{M} \cup \{M\}$; go to **(S2)**;
otherwise go to **(S2)**;
otherwise take $\delta_i = (\alpha : \beta)/\gamma$ and put $i := i + 1$;

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then $J := J \cup \{\beta\}$; $M := M \cap \mathrm{Mod}(\gamma)$; go to **(S6)**;

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then $J := J \cup \{\beta\}$; $M := M \cap \text{Mod}(\gamma)$; go to **(S6)**;
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(S6) If (M, J) is an MJ–pair then go to **(S4)**; otherwise go to **(S2)**.

Transition network

The algorithm for computing extensions of $T = (A, \Delta)$ can be depicted by a ***transition network*** $N = (V, E)$ such that

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- If a node v is labeled by (M, J) and $\delta = (\alpha : \beta)/\gamma$ is such that $M \subseteq Mod(\alpha)$ and $M \cap Mod(\neg\beta) = \emptyset$, then there is an arc labeled by δ leading to v' labeled by $(M \cap Mod(\gamma), J \cup \{\beta\})$.

Transition network (cont.)

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- A leaf is either
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Example 13.7

$T = (A, \Delta)$, where $A = \{p, q\}$, $\Delta = \left\{ \delta_1 = \dfrac{p : r \wedge \neg s}{r}\ ,\ \delta_2 = \dfrac{q : s}{s} \right\}$.

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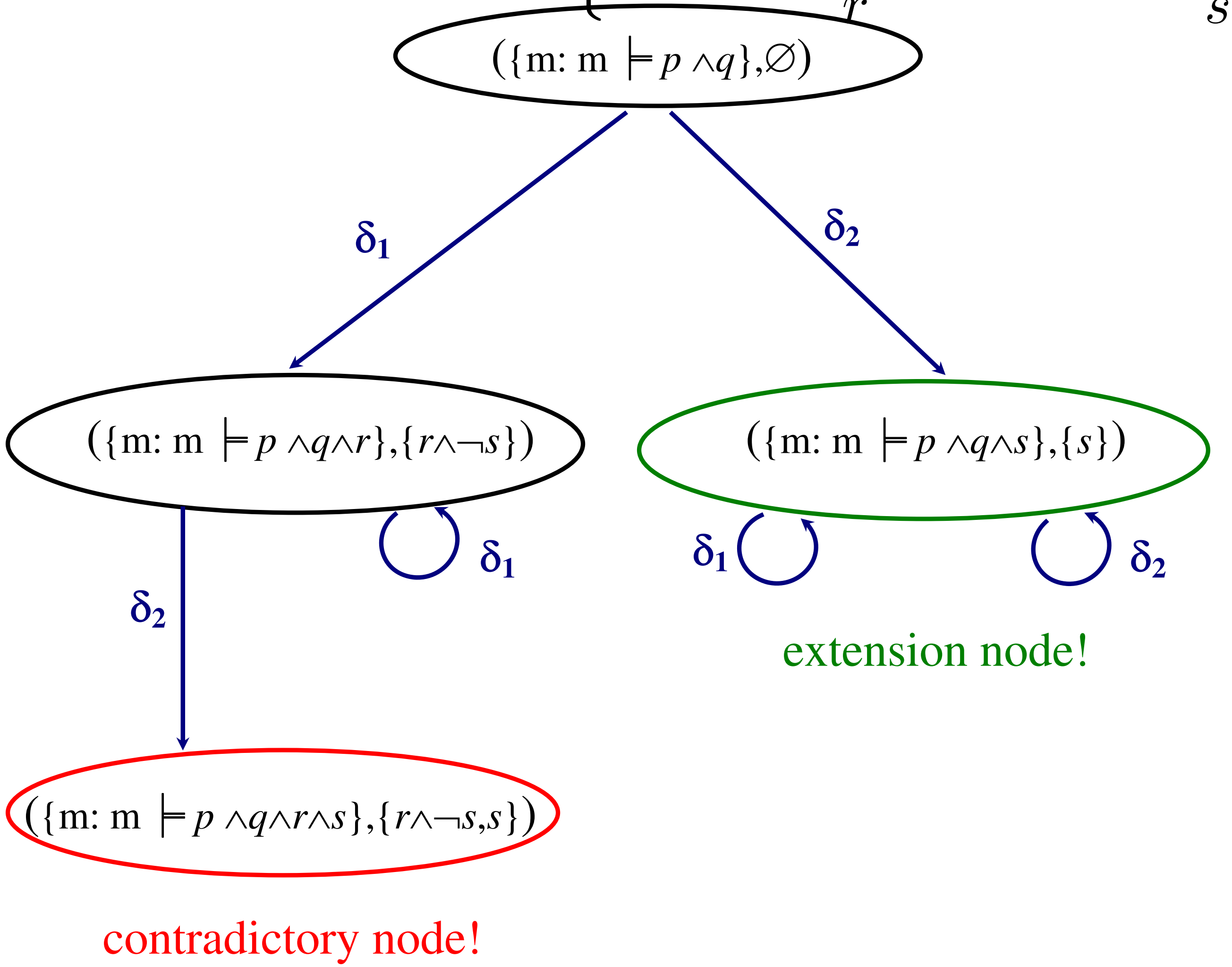

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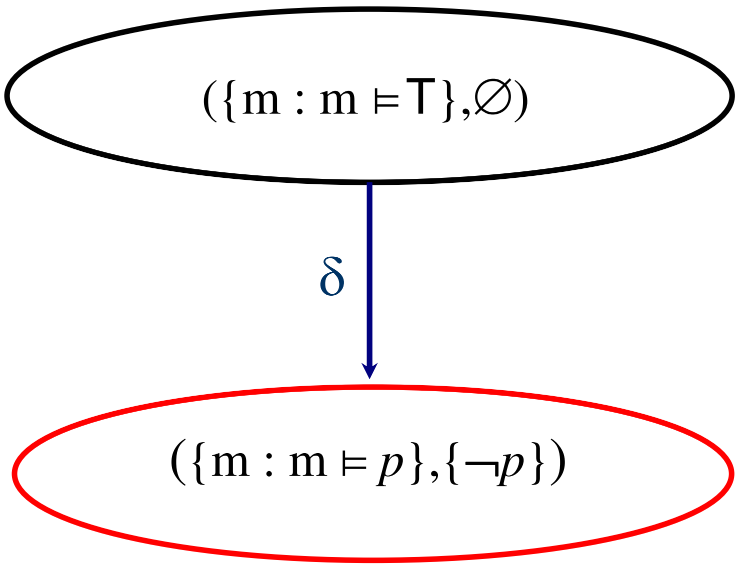

contradictory node!

Example 13.9

$T = (A, \Delta)$, where $A = \{p\}$ and $\Delta = \left\{ \delta_1 = \frac{p : q}{q}\ ,\ \delta_2 = \frac{p : q, \neg q}{r} \right\}$

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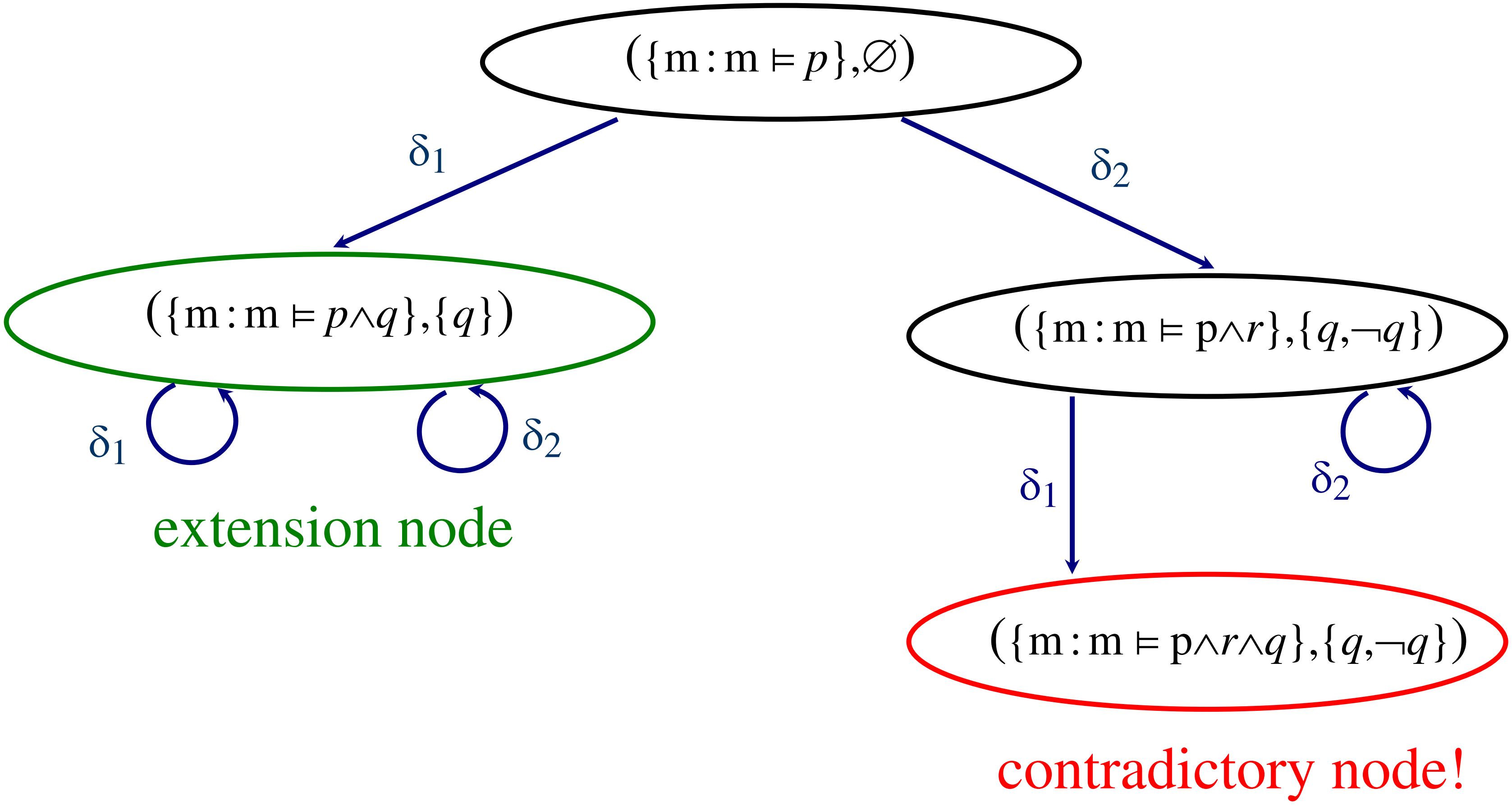

Problems with DL

Some default theories have no extensions, because the criterion of defaults' applicability is too weak — it enforces the default to be applicable, but after its application the default is not applicable.

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$$A = \emptyset, \quad \Delta = \left\{ \frac{: p}{\neg p} \right\}.$$

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- The consequent of the default, together with axioms and consequents of previously applied defaults, denies some justification of previously applied defaults.
  For example,
  $$A = \emptyset, \quad \Delta = \left\{ \frac{: p \wedge r}{p}, \frac{p : \neg r}{\neg r} \right\}.$$
- The consequent of the default contradicts some sentence derivable from axioms and consequents of already applied defaults.
  For example,
  $$A = \{p \rightarrow q\}, \quad \Delta = \left\{ \frac{: p}{p}, \frac{p : s}{\neg q} \right\}.$$

Two default logics

In order to overcome these problems, the alternative default system was introduced (Łukaszewicz 1988). Henceforth we will use the following terminology:

- Reiter's default logic (RDL) for the standard default logic
- Alternative default logic (ADL) for the new one.

Alternative Default Logic

Alternative extension

Let $T = (A, \Delta)$ be a closed default theory over the language $\mathcal{L}$. Define two operators Γ_1 and Γ_2 specified for pairs of sentences such that for any pair (B, J) , $\Gamma_1(B, J)$ and $\Gamma_2(B, J)$ are the smallest sets of sentences satisfying:

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A set $E \subseteq \mathcal{L}$ of sentences is an *alternative extension of T wrt F* iff $E = \Gamma_1(E, F)$ and $F = \Gamma_2(E, F)$. E is an ***alternative extension of*** T iff there is a set $F \subseteq \mathcal{L}$ of sentences such that E is an alternative extension of T wrt F.

Alternative extension (cont.)

Given two sets of sentences, B and J, if

then $\gamma \in \Gamma_1(B, J)$ and $\beta_1, \ldots, \beta_n \in \Gamma_2(B, J)$.

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Algorithm for computing alternative extensions

Algorithm

S1. $P := PERM(\Delta);$

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if (M, J) is Δ–stable then $\mathfrak{M} := \mathfrak{M} \cup \{M\}$; go to **S2**;

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S5. If $M \subseteq Mod(\alpha)$ and $M \cap Mod(\gamma) \cap Mod(\varphi) \neq \emptyset$ for every $\varphi \in J \cup \{\beta\}$ then $J := J \cup \{\beta\}$; $M := M \cap \text{Mod}(\gamma)$; go to **S4**;
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otherwise go to **S4**

Remark: (M, J) detemine in **S5** are MJ–pairs!

Example 13.10

Consider the following story:

- *On Sundays Bill usually goes fishing except when he is tired.*
- *If Bill worked hard the day before, then he is usually tired except when he woke up late.*
- *If Bill is on vacations, then he usually wakes up late except when he goes fishing.*
- *Today is Sunday, Bill worked hard yesterday, and he has vacations.*

Representation in DL:

$$A = \{s, w, v\}\,, \quad \Delta =$$

$$\left\{\delta_1 = \frac{s : g \wedge \neg t}{g}\,,\ \delta_2 = \frac{w : t \wedge \neg l}{t}\,,\ \delta_3 = \frac{v : l \wedge \neg g}{l}\right\}.$$

Example 13.10 (cont.)

$$A = \{s, w, v\}\,,\ \Delta = \left\{\delta_1 = \frac{s : g \wedge \neg t}{g}, \delta_2 = \frac{w : t \wedge \neg l}{t}, \delta_3 = \frac{v : l \wedge \neg g}{l}\right\}.$$

Take $(\delta_1, \delta_2, \delta_3)$.

Example 13.10 (cont.)

$$A = \{s, w, v\}\,,\;\; \Delta = \left\{\delta_1 = \frac{s : g \wedge \neg t}{g}, \delta_2 = \frac{w : t \wedge \neg l}{t}, \delta_3 = \frac{v : l \wedge \neg g}{l}\right\}.$$

Take $(\delta_1, \delta_2, \delta_3)$.

- δ_1 is applicable, so

$$M_1 = \left\{\begin{matrix} \{s, w, v, g, t, l\} \\ \{s, w, v, g, t, \neg l\} \\ \{s, w, v, g, \neg t, l\} \\ \{s, w, v, g, \neg t, \neg l\} \end{matrix}\right\},\;\; J_1 = \{g \wedge \neg t\},\;\; B = B_1 = Th(A \cup \{g\}).$$

Example 13.10 (cont.)

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- δ_2 is not applicable (t contradicts the justification of δ_1), so
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- δ_3 is not applicable (as in Reiter's criterion), so $M_3 = M_2, \ J_3 = J_2, \ B_3 = B_2$.

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Hence B is the alternative extension of T.

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Hence B' is another alternative extension of T.

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- δ_3 is applicable, so

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- δ_1 is not applicable (g contradicts the justification of δ_3), so $M_2 = M_1,\ J_2 = J_1,\ B_2 = B_1.$
- δ_2 is not applicable (as in Reiter's criterion), so $M_3 = M_2,\ J_3 = J_2,\ B_3 = B_2.$

Hence B'' is yet another alternative extension of T.

Example 13.10 (cont.)

- For the remaining sequences we obtain the same alternative extensions.

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- For the remaining sequences we obtain the same alternative extensions.
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  - $E_1 = Th(A \cup \{g\})$
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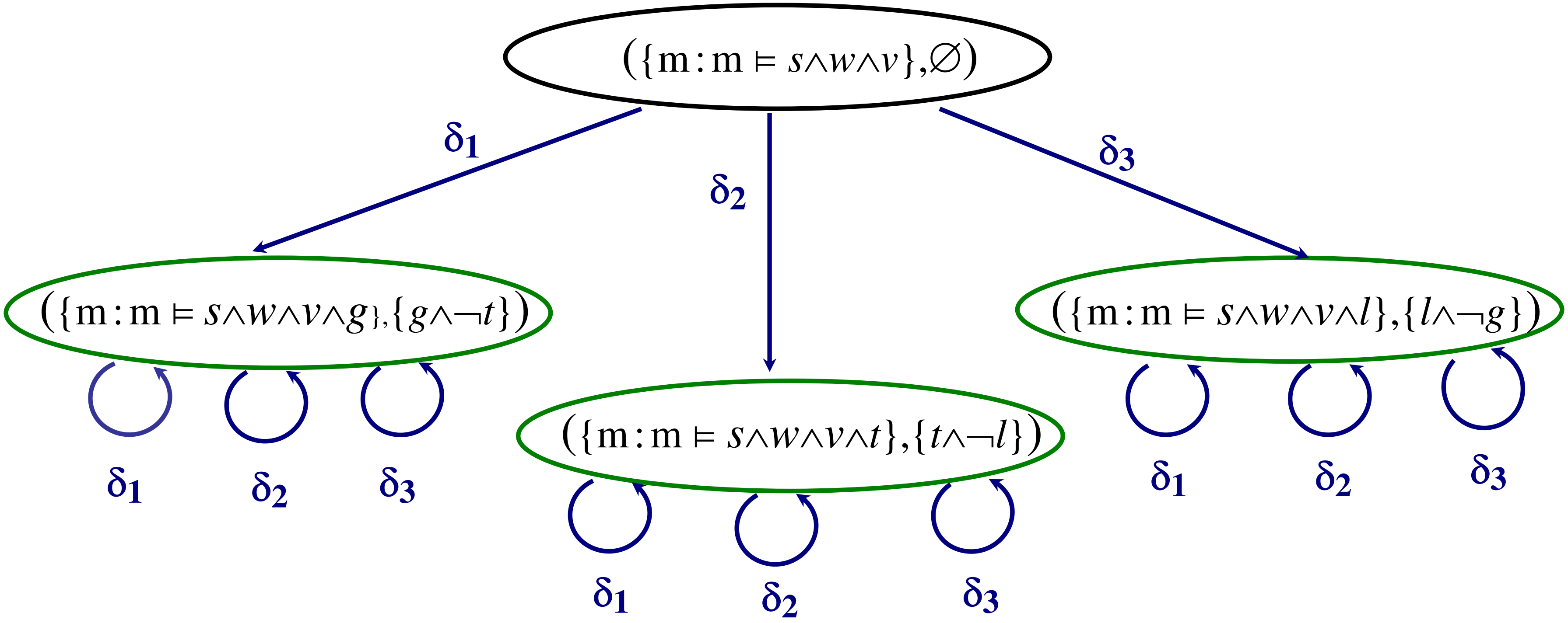
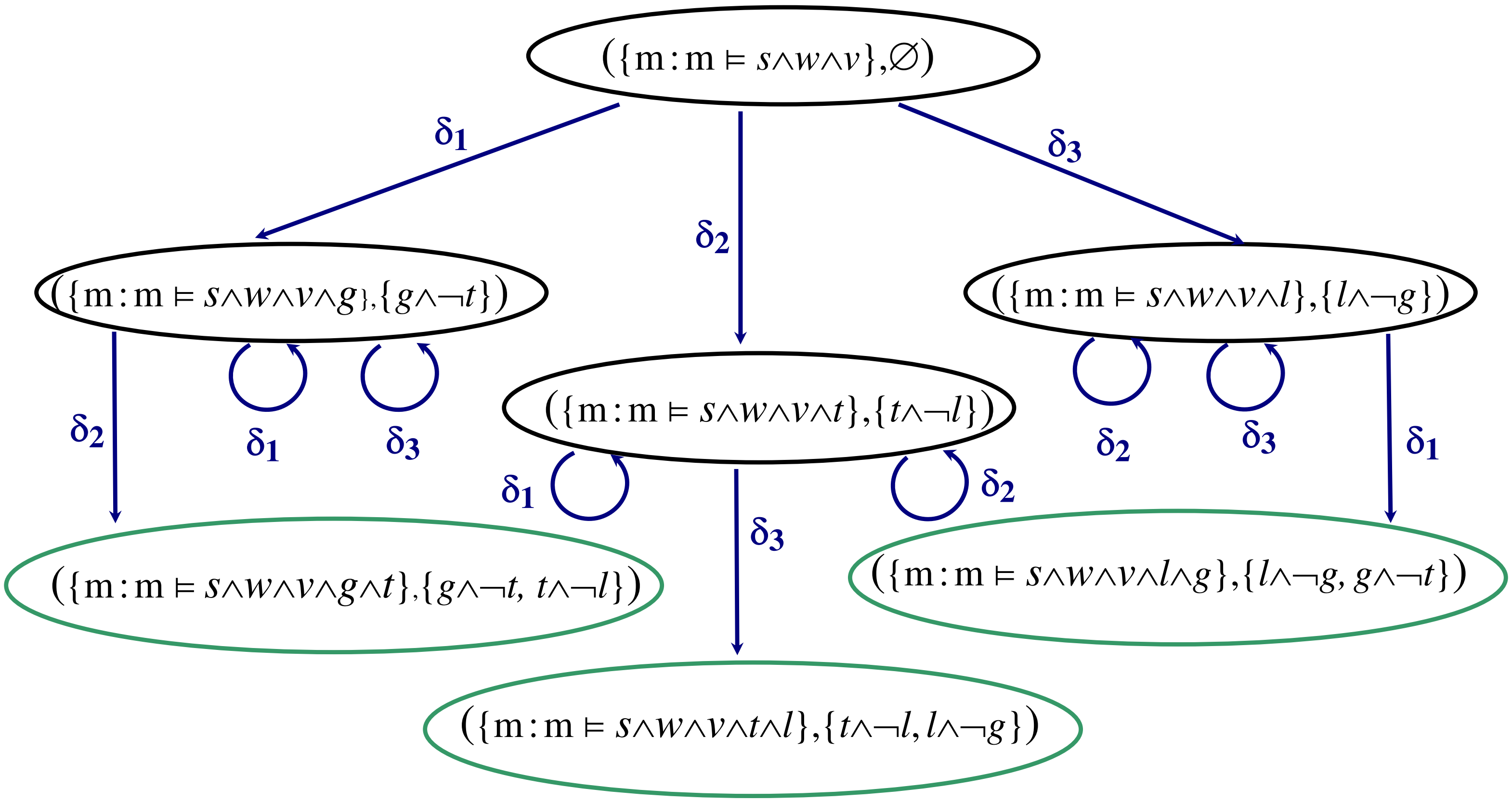
Example 13.10 (cont.)



Comparison with Reiter's DL

$$A = \{s, w, v\}\ , \Delta = \left\{\delta_1 = \frac{s : g \wedge \neg t}{g}, \delta_2 = \frac{w : t \wedge \neg l}{t}, \delta_3 = \frac{v : l \wedge \neg g}{l}\right\}.$$

Recall that T has no extension in RDL!



Example 13.11

Alternative extensions need not be maximal set of sentences. For example, consider the following default theory:

$$A = \{p\}\,, \quad \Delta = \left\{ \frac{:r}{p}\,, \frac{:q}{\neg r} \right\}.$$

One can easily check that T has two extensions:

$$E_1 = Th(\{p\}), \quad E_2 = Th(\{p, \neg r\})$$

wrt $F_1 = \{r\}$ and $F_2 = \{q\}$, respectively.

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Alternative extensions need not be maximal set of sentences. For example, consider the following default theory:

$$A = \{p\}\,, \quad \Delta = \left\{ \frac{:r}{p}\,, \frac{:q}{\neg r} \right\}.$$

One can easily check that T has two extensions:

$$E_1 = Th(\{p\}), \quad E_2 = Th(\{p, \neg r\})$$

wrt $F_1 = \{r\}$ and $F_2 = \{q\}$, respectively.

Non–maximality of alternative extensions is the result of applying a default which consequent is already believed, but which justification blocks the application of other defaults.

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- This, however, is not an important problem, since we are actually interested in the following problem:

  Given a default theory and a set of sentences determine whether all these sentences can be simultaneously believed, i.e., whether this is a subset of some (alternative) extension.

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  Let $T = (A, \Delta)$be a closed default theory and let E and E' be alternative extensions of T wrt F and F', respectively, such that $E \subseteq E'$ and $F \subseteq F'$. Then $E = E'$ and $F = F'$.

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- ***Semi–monotonicity***

  Let Δ_1 and Δ_2 be two sets of closed defaults such that $\Delta_1 \subseteq \Delta_2$ and let E_1 be an alternative extension of $T_1 = (A, \Delta_1)$ wrt F_1. Then $T_2 = (A, \Delta_2)$ has an alternative extension E_2 wrt F_2 such that $E_1 \subseteq E_2$ and $F_1 \subseteq F_2$.

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- A default theory (closed or open) $T = (A, \Delta)$ has an inconsistent alternative extension iff A is inconsistent.

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- The set of justifications for the inconsistent alternative extension is the empty set.
- If E is an alternative extension of a closed default theory T wrt F, then for every sentence $\varphi \in F$ it holds $E \not\vdash \neg\varphi$.

Thank you for your attention!

Any questions are welcome.